PROGRAM:

AIES - 1.py - C:/Users/ABBIRAMY/Documents/AIES - 1.py (3.14.2)

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```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}
def dfs(graph, start, goal):
    stack = [[start]]
    visited = set()
    order = []
    while stack:
        path = stack.pop()
        node = path[-1]
        if node in visited:
            continue
        visited.add(node)
        order.append(node)
        if node == goal:
            return path, order
        for nxt in reversed(graph[node]):
            if nxt not in visited:
                stack.append(path + [nxt])
    return None, order
path, order = dfs(graph, 'A', 'F')
print("Visiting order:", order)
print("Path found    :", path)
```

OUTPUT:

IDLE Shell 3.14.2

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```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: C:/Users/ABBIRAMY/Documents/AIES - 1.py ================
Visiting order: ['A', 'B', 'D', 'E', 'F']
Path found    : ['A', 'B', 'E', 'F']
>>>
```